ES 335 / ES 678 · PROPOSAL

Project proposal

Prepare and submit your proposal directly in the classroom portal. Due October 15, 2026 at 11:59 PM IST.

Project record

Course	ES 678
Project title	How Small Can a Reliable Prediction Set Be? Conformity-Score Design and Class-Conditional Coverage in Conformal Classification
Expected tools	Python 3.11, PyTorch + torchvision (MPS on Apple Silicon), NumPy, SciPy, pandas, Matplotlib, Git/GitHub, and LaTeX
Project members	Sumit Kumar Jha (26210180)
Proposal status	Draft saved · revision 29

Title and expected tools are shared with your registration. Update those project details if needed.

Proposal · submitted on this page

Write in each field below. Your text saves automatically a moment after you stop typing, and you can press **Save draft** at any time, even with fields still empty. The page uses your registered project and team; do not enter names or roll numbers again.

Research question

What specific question will your project answer? State it in one or two precise sentences.

An ordinary image classifier gives a single label and gives no signal about when it is likely to be wrong. Conformal prediction addresses this: instead of one guess, it returns a set of candidate labels that is guaranteed, with a chosen probability, to contain the correct class. Concretely, for a chosen error level alpha, the set includes the true label at least 1-alpha of the time (for example, at least 90% of the time when alpha = 0.1), and this holds without retraining the classifier. This project asks a focused question for a fixed, already-trained image classifier: how does the way we assign a score to each candidate label shape (a) the size of these sets, since smaller sets are more

useful, and (b) whether the guarantee holds fairly for every class individually rather than only on average across all images? In particular, we ask whether adding a distance-based term - one that measures how far an image sits from the typical appearance of each class and therefore flags unusual or unfamiliar images - can produce sets that are both smaller and more evenly reliable across classes than the standard scoring methods used today.

1143 / 1200 characters · minimum 20 to submit

Problem and intended claim

Why does this problem matter? What will you predict or analyze, and what would count as a useful answer?

Conformal prediction turns any classifier into one that, instead of a single guess, returns a set of plausible labels containing the true one at least (1-alpha) of the time (for example, 90% for alpha = 0.1), with no retraining. The catch is that this promise is only an average: it is guaranteed on average over the whole set of images, and can therefore hide the fact that the model covers some classes well and others badly. This matters whenever a wrong prediction carries a cost - for instance in medical triage or autonomous systems - because a set that looks reliable overall may still fail systematically for a minority class. We will predict, for each test image, a set of plausible labels, and analyse the trade-off between efficiency (keeping sets small, so a human or a downstream system can actually act on them) and fairness of coverage (making sure the guarantee is not quietly concentrated on the easy classes). A useful answer is a scoring method that keeps the overall guarantee, produces smaller sets on average, and narrows the gap between the best- and worst-covered class. A clear negative result - showing that distance information does not improve this trade-off - is equally valuable, and we will report it honestly rather than overselling it.

1268 / 1400 characters · minimum 30 to submit

Data plan

Name the source and access route; say what one row represents, the approximate number of observations, and the variables you expect to use.

All data are free and public and are downloaded automatically by the torchvision library, so there are no access or sharing restrictions. One row is one image together with its correct class label; the variables we use are the raw pixels and the classifier's predicted class probabilities. MNIST: 60,000 training and 10,000 test images of handwritten digits, 28x28 grayscale, 10 classes. CIFAR-10 (the main dataset): 50,000 training and 10,000 test 32x32 colour images across 10 classes. To test behaviour on unfamiliar images, we add out-of-distribution sets: FashionMNIST and KMNIST (10,000 test images each) against MNIST, and SVHN and CIFAR-100 (at least 10,000 each) against CIFAR-10. In total about 150,000 images. Preprocessing: we normalise pixel values using statistics computed from the training set only, and apply random crop and horizontal flip augmentation during training only, so nothing from the evaluation leaks into training. We check class balance, pixel-value ranges and near-duplicate images. MNIST is easy enough that the sets often shrink to a single label, so the main study of set size uses CIFAR-10, while MNIST serves only as a simple sanity check that the method is correct.

1203 / 1800 characters · minimum 40 to submit

Baseline and candidate methods

Name a simple baseline and the candidate method(s). Briefly explain why each is appropriate for the data and question.

Baselines: (1) the usual single-label prediction, included only as a reference point; (2) a simple logistic regression on MNIST, to confirm that a neural network is genuinely worth using; and (3) the naive score, which ranks a label by one minus its predicted probability - valid but not adaptive, because it treats every image the same way. Candidate methods: adaptive prediction sets (APS) rank labels by how much probability sits above them, so the set automatically adjusts to each image; regularised APS (RAPS) adds a penalty for unusually large sets. The proposed score combines two well-known ideas. The first is temperature scaling, a standard trick that makes predicted probabilities more realistic by softening overconfident outputs. The second is a Mahalanobis distance, which measures how far an image's internal features lie from the typical

features of each class, so unusual images are treated with more caution. The score adds a distance term and a calibrated-probability term together; when the distance weight is set to zero, the score falls back to temperature scaling alone, which gives us a clean comparison. A distance term helps because unfamiliar images sit far from the class centres and should therefore receive larger sets.

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Evaluation plan

Describe the split or validation approach, the metric(s) and why they fit, and how you will prevent leakage while tuning.

We split the data four ways before running any comparison, so no choice is made after seeing the results: a training set (to fit the classifier), a design set (to tune every setting - the temperature, the distance weight and the RAPS parameters - and to choose the method), a calibration set (used only to compute the cutoff that defines the sets), and a test set (opened once, for the final numbers). This prevents leakage: tuning uses the design set only; the class centres and spread that the distance relies on are also measured on the design set, never on the training data; and the cutoff comes only from the calibration set. Metrics: (i) how often the true label actually falls inside the set, checked against the promised level; (ii) average, median and worst-case set size, where smaller is better; (iii) how often sets contain exactly one label, or none at all; (iv) coverage for each class separately, and the gap between the best- and worst-covered class; (v) coverage across different confidence levels; and (vi) coverage and size on the unfamiliar images. We test error levels alpha of 0.01, 0.05, 0.10 and 0.20, repeat every run over 3-5 random seeds, and report averages together with their spread so the results are not read from a single lucky split.

1268 / 1600 characters · minimum 40 to submit

Likely challenge

Identify one realistic data, modeling, evaluation, access, or privacy challenge and how you will handle it.

Perfect per-class coverage is provably impossible for every possible input, so an overall (average) guarantee can still hide weak performance on particular classes - which is exactly what this project sets out to measure rather than to eliminate. Two further risks stand out. First, leakage: if we tune the scoring method on the same data used to set the cutoff that defines the sets, the guarantee breaks silently, so we keep the design set (used for tuning) and the calibration set (used for the cutoff) strictly separate. Second, the distance term may simply not help on CIFAR-10; we treat that as an anticipated outcome to report honestly rather than overclaim. We also use several random seeds and report error bars, keep a backup classifier in case the main one proves unstable, and lead with CIFAR-10 while treating MNIST only as a simple sanity check.

859 / 900 characters · minimum 15 to submit

Track-specific plan

ES 335: describe the useful interaction your website will provide. ES 678: describe the graduate-level depth analysis you plan to include.

Graduate-level depth: a study of the trade-off between small sets and fair per-class coverage, built on the adaptive prediction sets (APS) method of Romano et al. (2020) and extended with a temperature-scaled, Mahalanobis-distance score. The theoretical angle is the efficiency-versus-conditional-coverage frontier: how small sets can get before per-class coverage degrades. Four analyses: (1) Theory - a proof that coverage lies between 1-alpha and 1-alpha + 1/(n+1) for n calibration points, explaining this finite-sample gap. (2) Calibration and uncertainty - per-class coverage, the best-to-worst gap, and a comparison against Mondrian conformal prediction (per-class, with its own guarantee) and temperature scaling alone. (3) Sensitivity and ablation - vary alpha and the distance weight, and build the score up step by step (naive, APS, temperature, distance) to isolate each contribution. (4) Failure audit - which classes and inputs break coverage, behaviour on unfamiliar images where the guarantee fails, and how a small calibration set widens the gap. This covers four addendum options: derivation, calibration/uncertainty, sensitivity/ablation, and failure audit.

1176 / 1200 characters · minimum 20 to submit

ES 335 pairs have one shared proposal record; either confirmed member may update it, and the page tells you if your partner has saved something newer. ES 678 students submit individually. Saving a draft and later submitting, or editing after submitting, updates the same proposal until the deadline.

Restored text you had typed on this device at 12:57 PM that had not reached the server. It will save automatically.

Save draft Submit proposal

All changes saved at 01:03 PM IST